

Network Science HS23

Assignment 1

Blockchain & Distributed Ledger Technologies Group

FOR STUDIES PURPOSES ONLY

Instructions:

- You must submit your answers through OLAT.
- You have three weeks to submit the results. The deadline is 2023.10.23, 23:59 (Zürich time).
- Answer means
 - Code or theoretical solution, depending on the exercise request.
 - A document with the results clearly visible.
 - Possible formats are PDF, Python Notebooks.
 - All code must be already executed and results clearly visible.
- Show all your work and reasoning clearly.
- You can do this assignment as a group of up to two persons.
- It should be clear which answer belongs to which question.
- In every file you submit, indicate clearly your name and matriculation number.
- The maximum number of points you can achieve in this assignment is 30.
- The grade will be computed at the end of the course, over the total number of points achieved in all the assignments.

1 Permutations and Irreducibility (6 points)

- Definition 1.** An $n \times n$ elementary permutation matrix, denoted as $P^{(i,j)}$, is a square matrix obtained from the $n \times n$ identity matrix (denoted as I) by swapping the positions of exactly the two rows i and j .

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{bmatrix} a & b & c \\ g & h & i \\ d & e & f \end{bmatrix}$$

Figure 1: Elementary Permutation Matrix $P^{(2,3)}$ left-multiplication.

Show that the multiplication of matrix $P = P^{(i,j)}$ by a vector v has the effect of switching entries i and j of v . Show also that the left multiplication of matrix P by an $n \times m$ matrix A has the effect of switching rows i and j of matrix A , while right multiplication of $m \times n$ matrix B by the same matrix has the effect of switching columns i and j .

- Definition 2.** A permutation matrix is a square matrix obtained from the same size identity matrix by a permutation of rows. Every row and column of a permutation matrix contains precisely one non-zero entry equal to 1. There are two 2×2 permutation matrices. There are six 3×3 permutation matrices. There are $n!$ $n \times n$ permutation matrices.

Prove that for any permutation matrix P we have $P^T = P^{-1}$.

Hint: Prove the thesis first on elementary permutation matrices, then use the fact that any permutation matrix can be written as a product of elementary permutation matrices.

- Definition 3.** A matrix A is said to be reducible if there exists an $n \times n$ permutation matrix P such that:

$$P^T A P = \begin{bmatrix} A_{1,1} & A_{1,2} \\ 0 & A_{2,2} \end{bmatrix}$$

with $A_{1,1}$ and $A_{2,2}$ square matrices. Otherwise the matrix is irreducible.

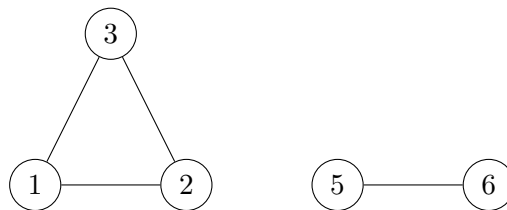


Figure 2: A disconnected graph of 5 nodes.

Consider the graph depicted in figure 2, show that the corresponding adjacency matrix is reducible. *Hint: find the permutation (and permutation matrix) such that you can apply the definition.*

- In class we characterized the adjacency matrices of connected graphs as irreducible matrices. Formally this result is a consequence of the following theorem:

Theorem 1. An undirected (directed) graph is connected (strongly connected) if and only if its adjacency matrix A is irreducible.

One could use this result to check the connectedness of a graph by just looking at the irreducibility of the adjacency matrix. Unfortunately, in real world applications, where the network size becomes very large, it is not straightforward to decide whether a matrix is irreducible or not. Give a possible explanation of why def. 3 and theorem 1 do not provide a practical way to verify whether a graph is connected or not in real world examples.

5. **(bonus question)** When dealing with real-world networks, meaning large networks, a more operative solution to check irreducibility (thus connectedness) is provided by the following theorem:

Theorem 2. *A real non-negative $n \times n$ matrix A is irreducible if and only if $\forall i, j \in [1, \dots, n], \exists k = k(i, j) > 0$ such that $(A^k)_{ij} > 0$.*

Use theorem 1 to prove the following theorem. .

2 Clustering Coefficient - Coding Exercise (6 points)

For the provided datasets (you can find the dataset files in the Materials/Session 03 folder on OLAT):

1. Plot the average degree of the nearest neighbours k_{nn} as a function of the vertices degree.
2. Super impose the same quantity for the randomised network.

Hint: The randomised networks are obtained via multiple edge swaps via the `networkx` function `nx.algorithms.smallworld.random_reference`. Make sure to set the parameter `connectivity = False` to have faster execution. If execution is still slow, we provide randomised datasets.)

3. Compute the assortativity coefficient of the real network.
4. Compute the assortativity coefficient of the randomised network.

Datasets Provided

- `graph_celegansInteractome.gml`: C. elegans interactomes. Nodes represent proteins and Edges represent protein-protein interactions in *Caenorhabditis elegans* (nematode), Simonis and et al. [2009].
- `graph_AstroPh.gml`: AstroPhysics Arxiv collaborations. Nodes represent authors of papers submitted to arxiv.org in the AstroPh category and Edges represent co-authorship between two authors, Leskovec et al. [2007].
- `graph_CondMat.gml`: Condensed Matter Arxiv collaborations. Nodes represent authors of papers submitted to arxiv.org in the CondMat category and Edges represent co-authorship between two authors, Leskovec et al. [2007].
- `graph_chess.gml` Kaggle chess players: Nodes represent chess players and Edges represent chess match among the worlds top chess players, *Chess ratings - Elo versus the Rest of the World* [2010].
- `graph_facebook.gml`: Facebook friendships. Nodes represent Facebook users and Edges represent their friendship relations collected from survey participants, Leskovec and McAuley [2012].
- `graph_internet.gml`: Internet AS graph. Nodes represent autonomous systems and Edges represent connections between them, Karrer et al. [2014].

3 One-dimensional Lattice (9 points)

1. Consider a chain, defined as follows:

Definition 4. A chain $C = \{V, E\}$ of length n is an undirected one dimensional lattice with no periodic condition and coordination number $k = 1$, where $V = \{1, \dots, n\}$ and $E = \{(i, i+1), i = 1, \dots, n-1\}$.



Figure 3: A chain C of length n .

Write down the explicit expression of the distance $d_{i,j}$ between any two node i, j of a chain C of length n . Write down clearly all the passages and your reasoning.

2. Use the result from point 1 to calculate the explicit expression of the Average Distance of a chain C of length n . Remember the Average Distance is defined as:

$$\langle d_{i,j} \rangle = \frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n d_{i,j}.$$

Hint: Use the symmetry of the problem; especially the fact that $d_{i,j} = d_{j,i}$.

3. Consider a ring, defined as follows:

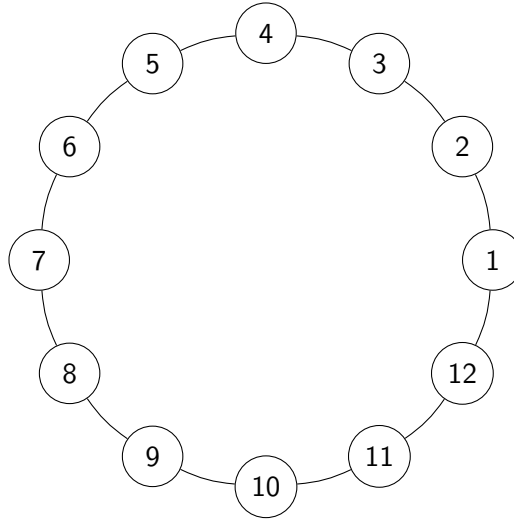


Figure 4: A ring of size $n = 12$

Definition 5. A ring $R = \{V, E\}$ of size n is an undirected one dimensional lattice with periodic condition and coordination number $k = 2$, where $V = \{1, \dots, n\}$ and $E = \{(1, 2), (2, 3), (3, 4), \dots, (n-1, n), (n, 1)\}$.

Write down the explicit expression of the Average Distance on a generic ring of size n when n is an odd number. Write down clearly all the passages and your reasoning.

Hint: take into account the symmetry of the problem. Start by finding the expression of L_i , the average

distance from a generic node i :

$$L_i = \sum_j d_{i,j}.$$

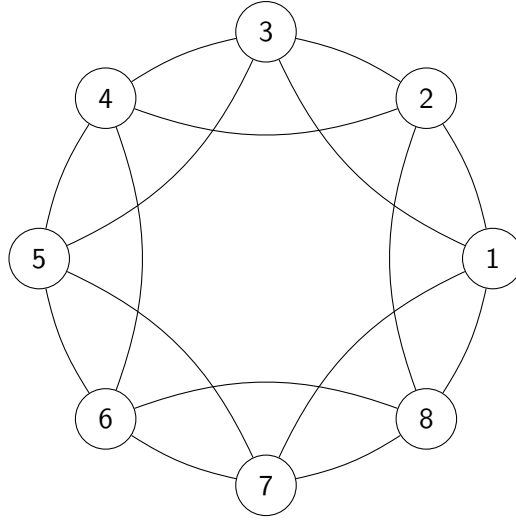


Figure 5: A 1-d lattice with periodic conditions of size $n = 8$ and $k = 2$.

4. Consider a generic 1-d lattice with coordination number k and number of nodes n , in the special case where n is odd and $\frac{n-1}{2k}$ is an integer (meaning $n-1$ is divisible by $2k$). Write down the explicit expression of the Average Distance of this network. Write down clearly all the passages and your reasoning.
5. **(bonus question)** Write down the explicit expression of the Average Distance for a 1-d lattice in the generic case where $n, k \in \mathbb{N}$, $n > k$.

4 Centralities - Coding Exercise (9 points)

For the provided datasets (you can find the dataset files in the Materials/Session 03 folder on OLAT):

1. Compute degree, closeness, betweenness and eigenvector centrality for each node.
2. Do a scatter plot of each pair of centrality (6 plots total). Compute Pearson, Spearman and Kendall correlation coefficient for each pair and note them on the scatter plots. *Hint: Centralities measures are available on `NetworkX`, while correlation coefficients are available on the module `scipy.stats`.*

For 2 vectors X and Y , the Pearson's r correlation is:

$$r(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y},$$

The spearman's ρ is:

$$\rho(X, Y) = r(\text{rank}(X), \text{rank}(Y)),$$

where $\text{rank}(X)$ is the sorted version of X . Kendall's τ is:

$$\tau(X, Y) = \frac{1}{\binom{N}{2}} \sum_{i>j} \text{sign}(X_i - X_j) \text{sign}(Y_i - Y_j),$$

where $\text{sign}(x)$ is 1 if $x > 0$, -1 if $x < 0$, and 0 if $x = 0$.

3. Repeat point 2 on randomised versions of the graphs provided. *Hint: Check point 2 of Exercise 2, for indications on the randomisation function to use in `NetworkX`.*
4. Briefly explain, for the Jazz collaboration network from Gleiser and Danon [2003], what each centrality measure means in practice.

Datasets provided

- `graph_eu_airlines.gml`: European airline network. Nodes represent airlines and Edges represent airline routes among European airports Cardillo et al. [2013]
- `graph_jazz_collab.gml`: Jazz collaboration network. Nodes represent jazz musicians and Edges represent collaborations in bands that performed between 1912 and 1940, Gleiser and Danon [2003].
- `graph_game_thrones.gml`: Game of Thrones co-appearances. Nodes represent Game of Thrones characters and Edges represent co-appearances of characters in the Game of Thrones series, Beveridge and Shan [2016].

References

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